

Bryan-Elliott Tam

+32 0472 13 14 28 | bryan_elliott_tam@protonmail.com | github.com/constraintAutomaton | linkedin.com/in/bryanelliotttam/ | constraint-automaton.pp.ua/ |  orcid.org/0000-0003-3467-9755

Software Engineer with experience in applied R&D, symbolic reasoning, knowledge representation, and web development. Seeking opportunities in software engineering, AI, and applied R&D. Personal projects and open-source contributions available at <https://constraint-automaton.pp.ua/project>.

SKILLS

- **Programming Languages:** TypeScript, Rust, Python, C++, Go, Prolog, SMT-LIB, Bash, SPARQL
- **Technologies:** Git, Jira, RDF, Svelte, Vue.js, React, Docker, MongoDB, Neo4j
- **Natural Languages:** French (Native), English (Fluent), Dutch (Basic)

WORK EXPERIENCES

Software Engineer (Decentralized Databases) September 2022 — Present
Ghent University – Imec *Gent, Belgium*

- Developed and implemented scalable solutions for querying decentralized knowledge graphs, integrated into the [Comunica](#) query engine, with associated peer-reviewed publications.
- Supported the Knowledge Graphs course and supervised master's and interns.
- Contributed to community engagement through the [SEMANTiCS 2025 Developers Workshop](#) committee and the [TREE W3C Community Group](#).

Research Software Engineer – Research Stay September 2025 — October 2025
Inria *Nice, France*

Developed optimization techniques for federated queries in the [Comunica](#) framework using RDF data shapes (FWO research stay).

Search Engine Developer – Contract May 2022 — September 2022
Université Laval *Sainte-Foy, Qc, Canada*

Developed a custom search engine to help architecture researchers retrieve relevant literature, implementing the backend in Go, the recommender system in Python, the frontend in JavaScript.

Localization Systems Developer – Contract May 2020 — September 2020
Université Laval *Sainte-Foy, Qc, Canada*

Transformed a 2D excavator localization system into a 3D solution, increasing positional accuracy by integrating map data and turret orientation; developed using C++ and Python.

IoT and Web Developer August 2018 — March 2020
Systèmes Vireo *Sainte-Foy, Qc, Canada*

- Led development and deployment of an end-to-end IoT platform for urban agriculture at a startup.
- Integrated embedded systems using C++, MQTT, and KiCad with full-stack web development technologies including React, TypeScript, Node.js, MongoDB, and Node-RED.
- Conducted on-site installation and testing of hardware and software components.
- Collaborated cross-functionally with other departments to ensure successful project delivery.

EDUCATION

Ghent University 2022 — Present
Doctorate, Computer Science Engineering *Gent, Belgium*

Université Laval 2020 — 2022
Master of Sciences, Computer Science *Sainte-Foy, Qc, Canada*

Université Laval 2015 — 2019
Bachelor of Engineering, Mechanical Engineering *Sainte-Foy, Qc, Canada*